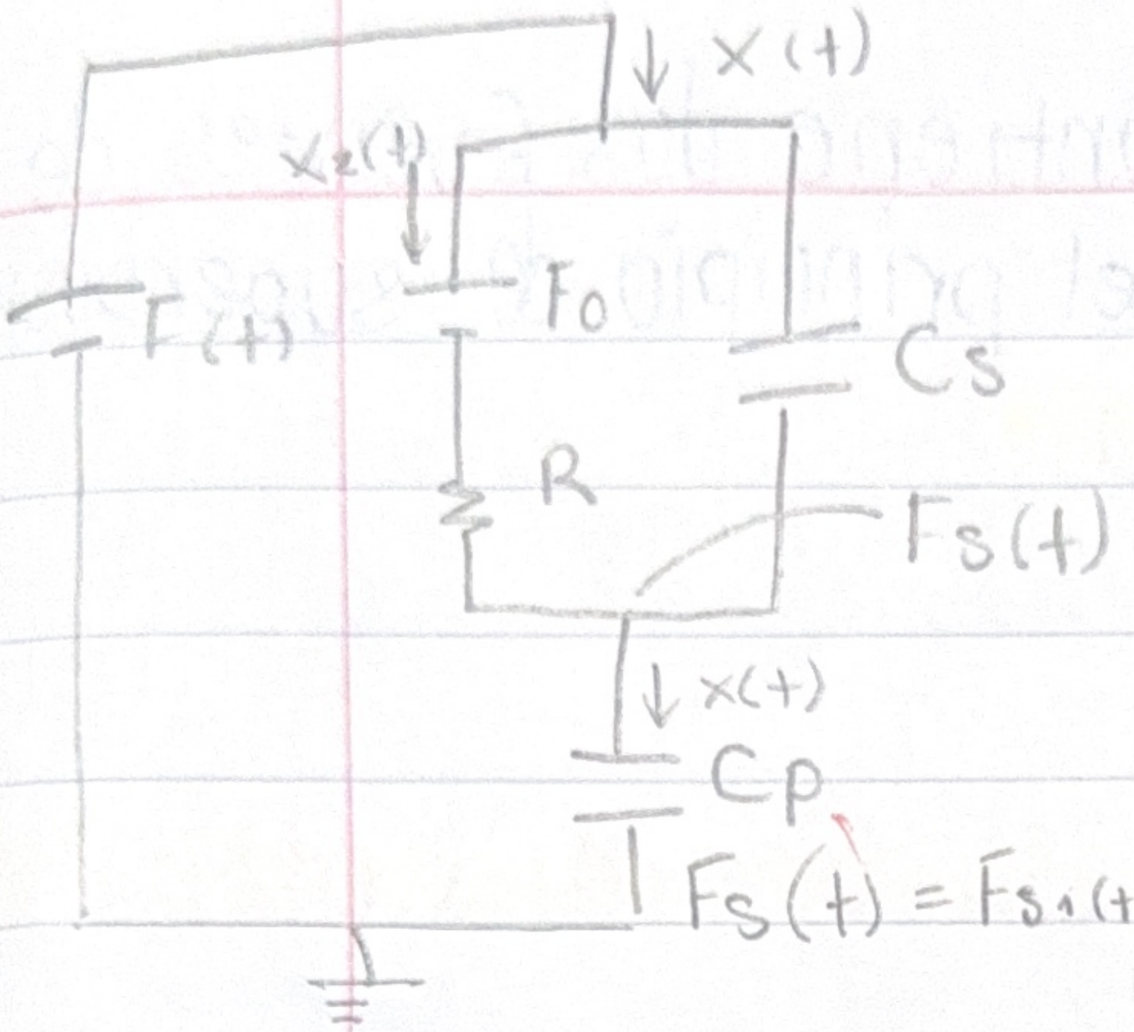
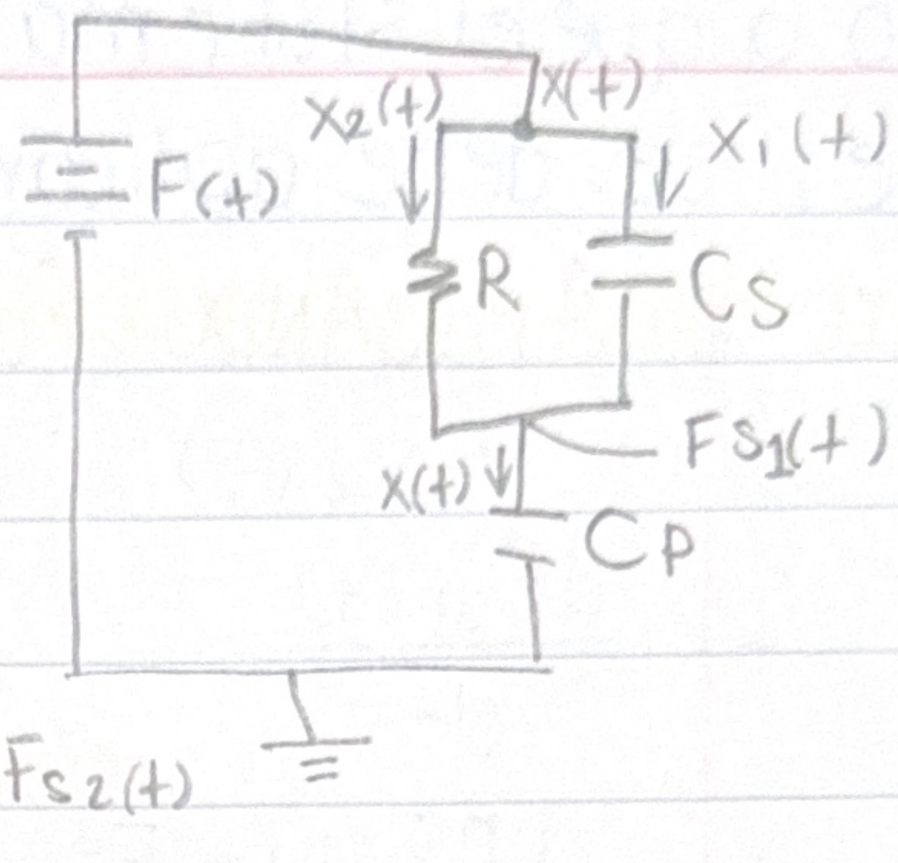
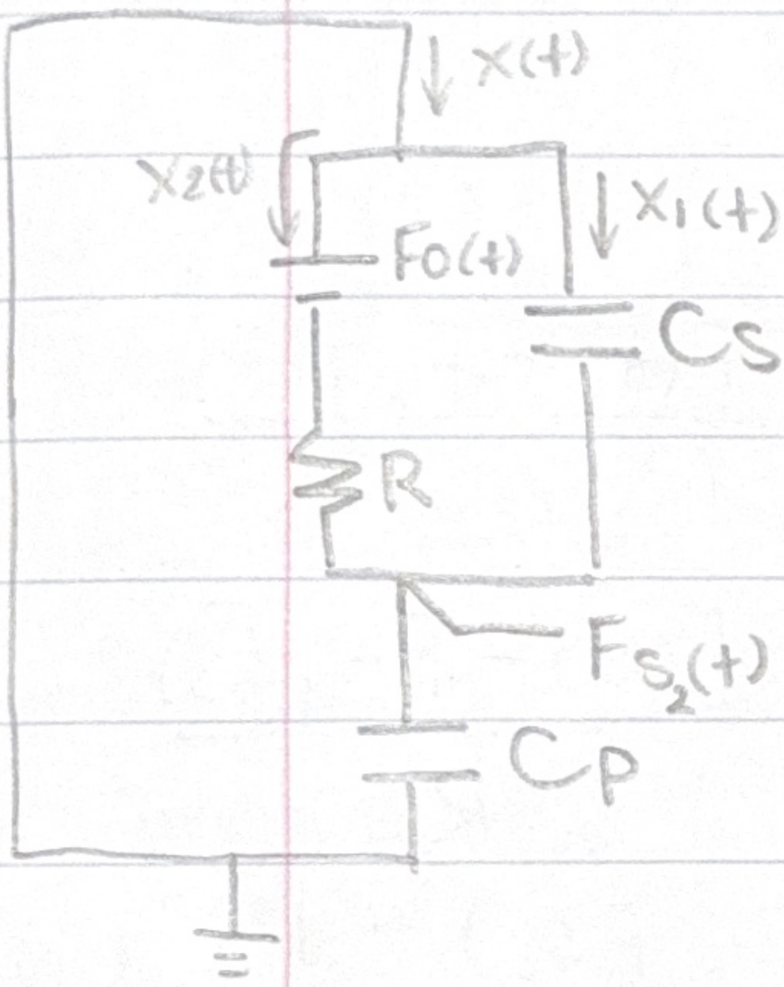
$x(t) = x_1(t) + x_2(t)$

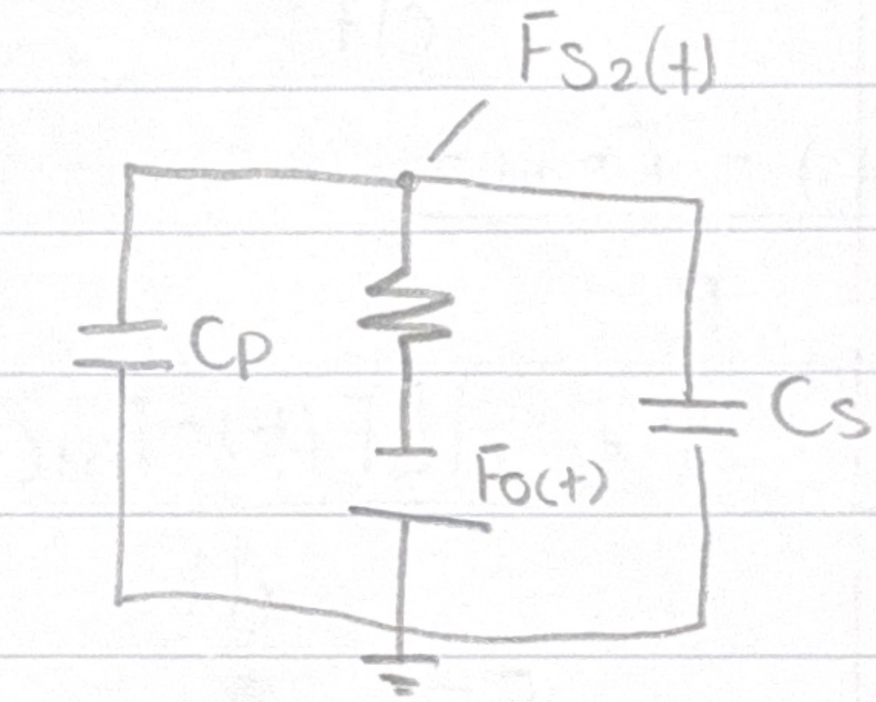
Apagando fuentes:



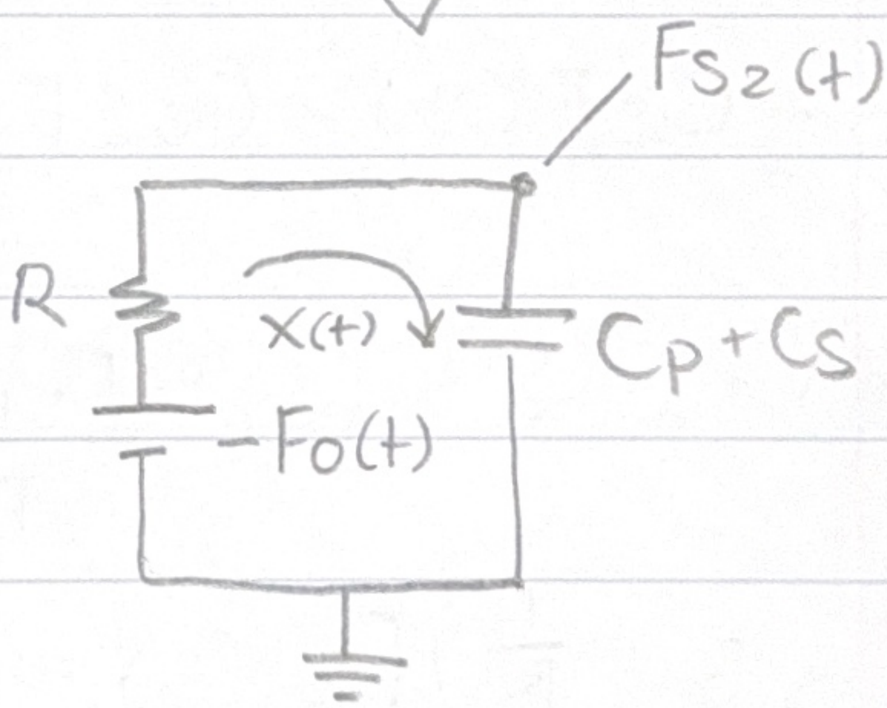




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ANALISIS MATEMATICO

Debido a que el sistema contiene dos fuentes de voltaje, se debe aplicar el principio de superposicion

CIRCUITO 1. APAGANDO $F_0(t)$

$$X(t) = X_1(t) + X_2(t)$$

$$X(t) = C_p \frac{dF_{s1}(t)}{dt}$$

$$X_1(t) = C_s \frac{d[F(t) - F_{S_1}(t)]}{dt}$$

$$X_2(t) = \frac{F(t) - F_{S_1}(t)}{R}$$

$$C_p \frac{dF_{S_1}(t)}{dt} = C_s \frac{d[F(t) - F_{S_1}(t)]}{dt} + \frac{F(t) - F_{S_1}(t)}{R}$$

$$C_p S F_{S_1}(S) = C_s S [F(S) - F_{S_1}(S)] + \frac{F(S) - F_{S_1}(S)}{R}$$

$$C_p S F_1(S) = C_s S F(S) - C_s S F_{S_1}(S) + \frac{F(S)}{R} - \frac{F_{S_1}(S)}{R}$$

$$C_p S F_{S_1}(S) + C_s S F_{S_1}(S) + \frac{F_{S_1}(S)}{R} = C_s S F(S) + \frac{F(S)}{R}$$

$$\left(C_p S + C_s S + \frac{1}{R}\right) F_{S_1}(S) = \left(C_s S + \frac{1}{R}\right) F(S)$$

$$\frac{R(C_p + C_s)S + 1}{R} F_{S_1}(S) = \frac{C_s R S + 1}{R} F(S)$$

$$F_{S_1}(S) = \frac{\frac{C_S R_S + 1}{R}}{\frac{R(C_P + C_S)S + 1}{R}} F(S)$$

$$F_{S_1}(S) = \frac{C_S R_S + 1}{R(C_P + C_S)S + 1} F(S)$$

CIRCUITO 2: APAGANDO $F(t)$

Recordando que $F_O(t) = \alpha F(t)$

$$-\alpha F(t) = R_x(t) + \frac{1}{C_P + C_S} \int x(t)\,dt$$

$$F_{S_2}(t) = \frac{1}{C_P + C_S} \int x(t)\,dt$$

$$-\alpha F(S) = R_x(S) + \frac{X(S)}{(C_P + C_S)S}$$

$$F_{S_2}(S) = \frac{X(S)}{(C_P + C_S)S}$$

$$F(S) = -\frac{1}{\alpha}\left(R + \frac{1}{(C_P + C_S)S}\right) X(S)$$

$$F(S) = -\frac{1}{\alpha} \frac{R(C_P + C_S)S + 1}{(C_P + C_S)S} X(S)$$

$$\frac{F_{S_2}(S)}{F(t)} = \frac{\frac{X(S)}{(C_P + C_S)S}}{-\frac{1}{\alpha}\frac{R(C_P + C_S)S + 1}{(C_P + C_S)S} X(S)} = -\frac{\alpha}{R(C_P + C_S)S + 1}$$

$$F_{S2}(S) = -\frac{\alpha F(S)}{R(C_P + C_S)S + 1}$$

FUNCION DE TRANSFERENCIA

$$F_S(S) = F_{S1}(S) + F_{S2}(S) = \frac{C_S R_S + 1}{R(C_P + (S)S + 1)} F_S - \frac{\alpha F(S)}{R(C_P + C_S)S + 1}$$

$$F_S(S) = \frac{C_S R_S + 1 - \alpha}{R(C_D + C_S)S + 1} F(S)$$

$$\frac{F_S(S)}{F(S)} = \frac{C_S R_S + 1 - \alpha}{R(C_P + C_S)S + 1}$$